

DECODED EXPLAINED

17 July, 2026

Topics

1. Corporate Average Fuel Efficiency (CAFE) Norms - **GS 3**
2. NITI Aayog roadmap for building India as a leading bioeconomy powerhouse- **GS 3**
3. The Supreme Court has directed all States and Union Territories to formulate a uniform policy for the premature release of elderly and terminally ill prisoners within three months.- **GS 2**
4. Reports emerged that a ransomware syndicate had leaked 14.3 GB of data containing 18,997 files explicitly linked to the Kudankulam Nuclear Power Plant (KKNPP) on the dark web.- **GS 3**

Corporate Average Fuel Efficiency (CAFE) Norms

CAFE norms are **government-regulated standards** that mandate automobile manufacturers to **meet specific fuel efficiency and emission targets** across their entire fleet of vehicles.

These regulations are implemented under the **Energy Conservation Act, 2001**

Comes under **Ministry of Power**

Objectives and Features

- To **lower greenhouse gas emissions**, especially CO₂
- To promote **energy-efficient** and **reduce India's dependence on crude oil** import.
- CAFE norms **apply to fleet-wide average emissions, not individual vehicles.**

The norms are implemented in phases (CAFE-1, CAFE-2, and now CAFE-3)

1. Introduced in **2017** (CAFE-1)
2. Strengthened under CAFE-2 (**2022** onwards)
3. The next phase, CAFE-3 is expected to be implemented from **April 2027.**

These norms form a crucial part of India's broader climate commitments, including achieving **net zero emissions by 2070**

BS norms vs CAFE norms

Bharat Stage (BS) Norms

- Reduce toxic tailpipe pollutants like **Particulate Matter** (complex mixture of extremely small solid particles and liquid droplets suspended in the air , eg- pm 10, pm2.5 etc..)
- Applies as a strict, maximum legal limit to **every individual vehicle** manufactured.
- **Objective** - Improve local air quality and public health by removing toxic pollutants.
- All two-wheelers, three-wheelers, and passenger/commercial vehicles.

CAFE Norms

- Reduce **carbon dioxide** (CO₂) emissions.
- Measured as a sales-weighted **average across an automaker's entire model lineup**.
- **Objective** - Improve national energy security and reduce overall fuel consumption .
- Passenger vehicles up to 3,500 kg (Petrol, Diesel, CNG, Hybrid, EV).

- BS4 Fuel: Contains more **sulphur** (50 parts per million), which creates more pollution when burned.
 - BS6 Fuel: Contains almost no **sulphur** (only 10 parts per million). It is 5 times cleaner.
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- BS4 Limit: Allowed to emit 250 mg/km of **nitrogen oxide**.
 - BS6 Limit: Only allowed to emit 80 mg/km of **nitrogen oxide**.

Context

NITI Aayog has recently announced the Roadmap for **Building India as a Leading BioEconomy Powerhouse by 2035.**

NITI Aayog

- **Apex Think Tank:** Replaced the Planning Commission on January 1, 2015, to design India's strategic policy framework.
- **Cooperative Federalism:** Drives a bottom-up approach by involving all state Chief Ministers in national policymaking.
- **PM-Led Structure:** Chaired by the Prime Minister, with an appointed Vice Chairperson and Chief Executive Officer.
- **Innovation Driver:** Spearheads national growth programs, including the **Atal Innovation Mission** for technology and entrepreneurship.

About Bioeconomy

- The **Department of Biotechnology (DBT)** defines bioeconomy as the **economic value** generated from the sustainable and innovative use of **biological resources** (including plants, animals, microorganisms and biomass), biological knowledge and biotechnology-based products and services.

Six National BioMissions

GeneIndia	Affordable gene and cell therapies (better treatments)
AgriBio 2.0	Climate-resilient gene-edited crops , biological farm inputs
BioX Foundry	Use tiny living organisms like bacteria to make useful products. , eg - eco-friendly plastics.
One Health Grid	Keep track of diseases in humans, animals, and the environment to stop outbreaks early
Marine Biotechnology	Benefits from the sea ,eg-Seaweed cultivation,
BioPharmaNext	Better medicines and vaccines production

Key Features of NITI Aayog's Biotechnology Roadmap

- ₹50,000 crore fund to **support new science and biotechnology projects.**
- **Production-Linked Incentives (PLI)** - Rewards companies that make biotechnology products in India.
- Supports better **medicines** and disease **testing** for healthier lives.
- **Industry-Academia Collaboration:** Scientists, colleges, and industries work together to solve real-life problems
- Makes approval of new medicines faster so people can get them sooner. **slash the current 900-day approval average down to competitive global standards.**

Major Challenges

- **Multi-agency overlaps** cause bottlenecks, pushing India's regulatory approval timelines to a staggering 900-day average, compared to approximately 500 days in competitor nations.
- **Limited Commercialisation of Research** - Many good ideas stay in laboratories instead of becoming useful products.
- Dependence on **Imported Technologies**
- **Talent and Infrastructure Constraints:** India hosts a critically low ratio of only **259 researchers per million people**, creating an extremely thin talent pipeline in advanced fields like bioinformatics and regulatory science

Wayforward

- Expanding **Public and Private R&D Investment**
- Strengthening **Biomanufacturing Infrastructure**
- **Simplifying Regulatory Processes**
- Building a **Skilled Biotechnology Workforce**

2019 PYQ

Atal Innovation Mission is set up under the

- (a) Department of Science and Technology
- (b) Ministry of Labour and Employment
- (c) NITI Aayog
- (d) Ministry of Skill Development and Entrepreneurship

Context

- The Supreme Court has directed all States and Union Territories to formulate a uniform policy for the premature release of elderly and terminally ill prisoners within three months.
- The judgment was delivered on a Public Interest Litigation (**PIL**) filed by the National Legal Services Authority (**NALSA**).
- The Court invoked its extraordinary powers under **Article 142** of the Constitution.

What is PIL ?

Public Interest Litigation (PIL) is a legal tool that allows any citizen or organization to file a lawsuit on behalf of the public, or on behalf of marginalized groups who cannot afford legal representation

What is Article 142 ?

Article 142 is a unique constitutional power that allows the Supreme Court of India to issue any order necessary to achieve complete justice when existing laws are silent or insufficient.

About National Legal Services Authority (NALSA)

- NALSA is a statutory authority established under the **Legal Services Authorities Act, 1987**, which came into force on 9 November 1995.
- **Objective:** It seeks to provide free and competent legal services to weaker and vulnerable sections and ensure equal access to justice under Article 39A of the Constitution.
- **Composition:** The **Chief Justice of India** is the **Patron-in-Chief**, while a **serving or retired Supreme Court Judge** serves as the **Executive Chairman**.

Who all are entitled to free legal services, under Legal Services Authorities Act

- A member of a **Scheduled Caste or Scheduled Tribe**
- A **victim of trafficking** in human beings or beggar as referred to in Article 23 of the Constitution
- A woman or a child
- **Transgender persons earning less than ₹4 Lakh per annum ,states have thresholds at ₹2 Lakh.**
- A **mentally ill** or otherwise **disabled** person;
- A person under circumstances of undeserved want such as being a **victim of a mass disaster**, ethnic violence, caste atrocity, flood, drought, earthquake or industrial disaster; or
- An **industrial workman**
- A person in receipt of annual income less than the amount mentioned in the following schedule (or any other higher amount as may be prescribed by the State Government), if the case is before a Court other than the Supreme Court, and **less than Rs 5 Lakh, if the case is before the Supreme Court. baseline limit is set at ₹ 1 Lakh by majority of states**

About State Legal Services Authorities (SLSAs)

- SLSAs are statutory bodies constituted in every State under the **Legal Services Authorities Act, 1987**.
- **Objective:** They implement NALSA's policies and programmes at the State level and ensure that eligible persons receive free legal aid and legal representation.
- **Composition:** The **Chief Justice of the High Court** serves as the **Patron-in-Chief**, while a **serving or retired High Court Judge** functions as the **Executive Chairman**.

Prelims 2020.

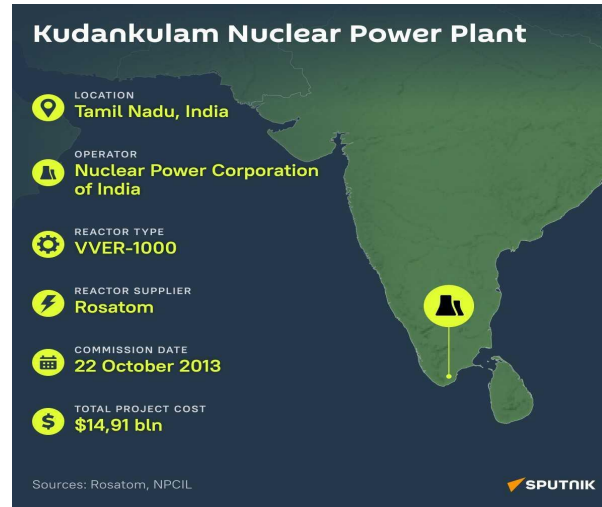
In India, Legal Services Authorities provide free legal services to which of the following type of citizens?

1. Person with an annual income of less than ₹ 1,00,000
2. Transgender with an annual income of less than ₹ 2,00,000
3. Member of Other Backward Classes (OBC) with an annual income of less than ₹ 3,00,000
4. All Senior Citizens

- A. 1 and 2
- B. 1 and 3
- C. 2 and 4
- D. 1,3 and 4

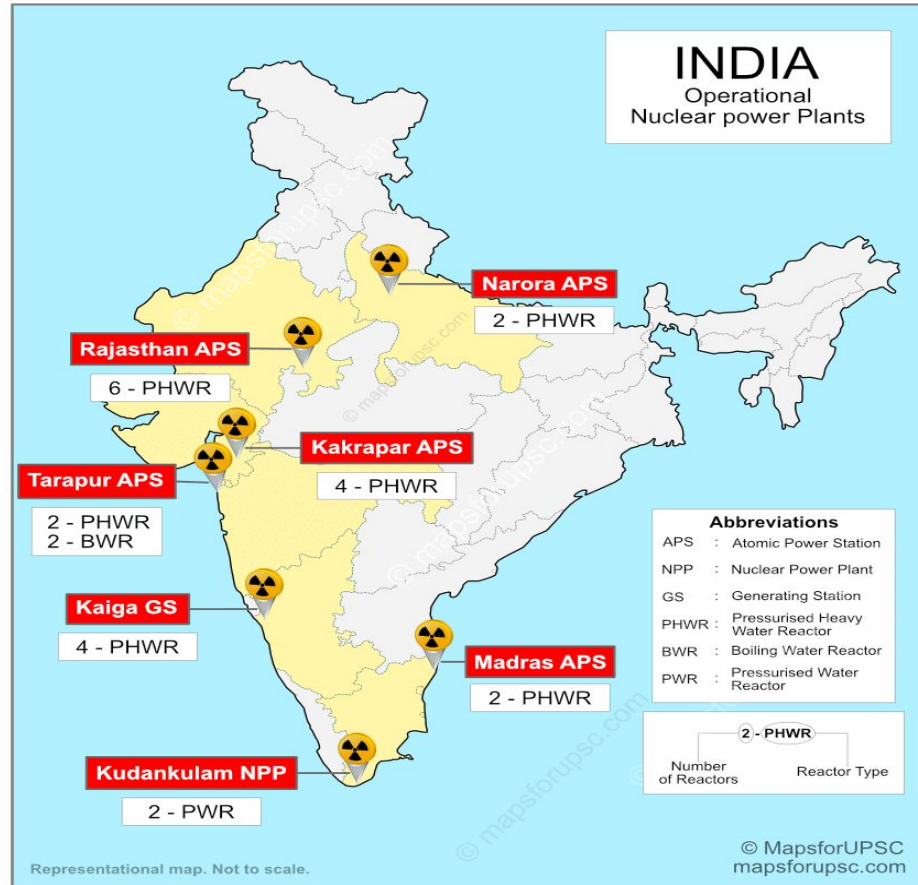
Context

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Nuclear power plants in India

- **Tamil Nadu** — Kudankulam Nuclear Power Station (KKNPP) & Madras Atomic Power Station (MAPS, Kalpakkam)
- **Maharashtra** — Tarapur Atomic Power Station (TAPS)
- **Gujarat** — Kakrapar Atomic Power Station (KAPS)
- **Rajasthan** — Rajasthan Atomic Power Station (RAPS, Rawatbhata)
- **Karnataka** — Kaiga Generating Station (KGS)
- **Uttar Pradesh** — Narora Atomic Power Station (NAPS)



THANK YOU